

Random variables, distributions and expectation

From single events to whole patterns.

THE IDEA

A **random variable** is a number whose value depends on chance — a die roll, a patient's blood pressure, a model's loss on the next batch.

Type	Meaning	Examples
Discrete	Countable values	Die roll, number of patients, clicks
Continuous	Measured values	Height, temperature, loss

EXPECTED VALUE

The long-run average of a random variable — each value weighted by its probability.

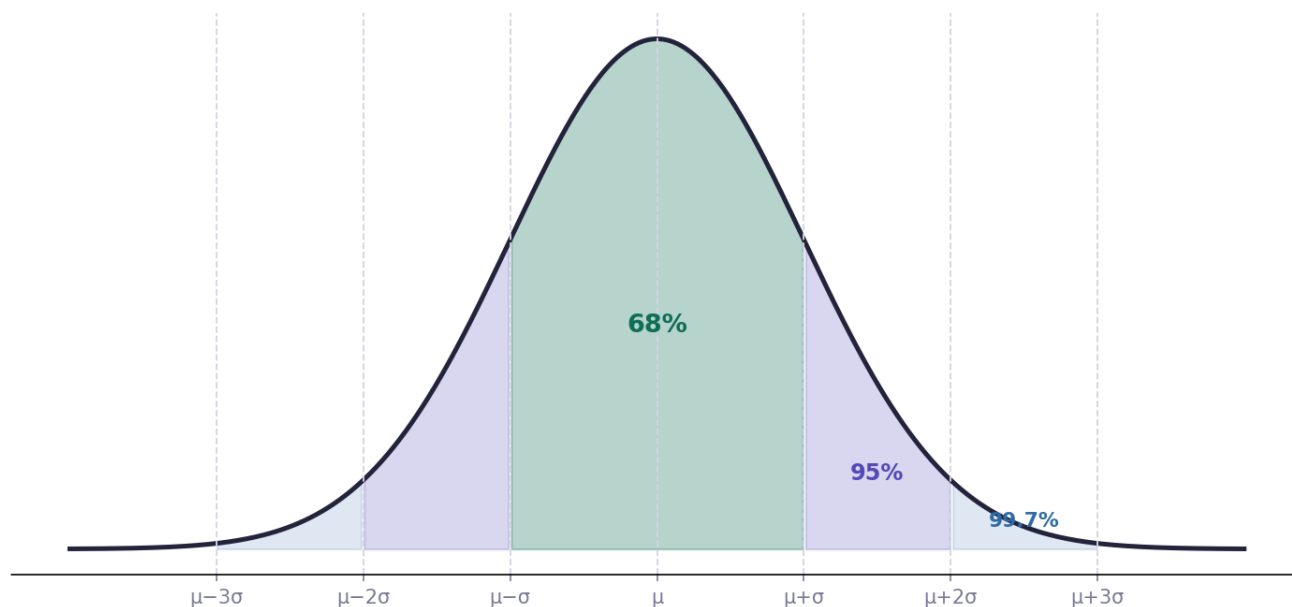
$$E[X] = \sum (\text{value} \times \text{its probability})$$

A fair die has $E[X] = 3.5$, which you can never actually roll. That is the point: the expected value is an average, not an outcome.

THE NORMAL DISTRIBUTION

The bell curve, described completely by two numbers: the mean μ (where the centre sits) and the standard deviation σ (how wide it spreads).

The normal distribution — the 68 · 95 · 99.7 rule



Within	Contains
$\mu \pm 1\sigma$	about 68% of the data
$\mu \pm 2\sigma$	about 95%

WORKED EXAMPLES

TOPIC 5 — RANDOM VARIABLES & DISTRIBUTIONS

from single events to whole patterns

RANDOM VARIABLE

a number whose value depends on chance.

DISCRETE — countable (a die roll, number of patients)

CONTINUOUS — measured (height, temperature, loss)

EXPECTED VALUE $E[X]$ — the long-run average

$$E[X] = \sum (\text{value} \times \text{its probability})$$

example : a fair die

each face has probability 1/6 :

$$(1+2+3+4+5+6) \times 1/6 = 21/6 = 3.5$$

you can never roll 3.5 — it is the AVERAGE, not an outcome

A SECOND EXAMPLE — a biased coin game

win 10 with probability 0.3, lose 5 with probability 0.7

$$E[X] = (10 \times 0.3) + (-5 \times 0.7)$$

$$= 3 - 3.5 = -0.5$$

negative → on average you lose 50 cents per game

THE NORMAL DISTRIBUTION

the bell curve. described by just two numbers :

mean μ — where the centre sitsstd σ — how wide the spread is

the 68 - 95 - 99.7 rule :

68% of data within 1σ of the mean95% within 2σ 99.7% within 3σ **ML LINK :** "expected loss" is literally $E[X]$. Weights are initialised from a normal distribution, and the 2σ rule is how outliers get detected.

Why it matters for ML: "expected loss" is literally $E[X]$ — the loss averaged over the data distribution. Neural network weights are initialised by drawing from a normal distribution, and the 2σ rule is a standard way to flag outliers.

KEY POINTS FROM THIS TOPIC

Random variable	a number decided by chance
Expected value	$E[X] = \sum (\text{value} \times \text{probability})$
Normal distribution	described by mean μ and std σ
68 · 95 · 99.7	within 1σ , 2σ , 3σ of the mean

TEST — Topic 5

1. A coin pays \$6 for heads and \$0 for tails. What is $E[X]$? $\rightarrow (6 \times 0.5) + (0 \times 0.5) = 3$
2. A game wins \$20 with probability 0.2 and loses \$4 with probability 0.8. Find $E[X]$ and say whether it is worth playing. $\rightarrow (20 \times 0.2) + (-4 \times 0.8) = 4 - 3.2 = 0.8$; **positive, so yes**
3. Heights are normal with $\mu = 170$ cm and $\sigma = 8$ cm. Between which two heights do about 95% of people fall? $\rightarrow \mu \pm 2\sigma = 170 \pm 16 \rightarrow 154 \text{ cm to } 186 \text{ cm}$